cognizant Flowsource

Proxied Authentication

using **Google Cloud Identity-Aware Proxy (IAP)**

with **Microsoft Entra ID (Azure Active Directory)**

To securely authenticate users accessing your applications, you can configure an Application Load Balancer (ALB) to manage authentication. This configuration delegates the responsibility of user authentication to the Load Balancer, allowing your applications to dedicate their resources to executing the core business logic.

This document outlines the steps to set up Single Sign-On (SSO) offloading using Google Cloud Identity-Aware Proxy (IAP) as the authentication proxy, in conjunction with Microsoft Entra ID (formerly Azure Active Directory), within a Kubernetes environment.

Prerequisites

- Kubernetes Cluster (GKE)
- Application Load Balancer (ALB)
- Microsoft Entra ID (Azure Active Directory)
- OAuth App in Microsoft Azure

Permissions required on GCP Service Account

- roles/identitytoolkit.viewer
- roles/iap.Admin
- roles/compute.networkViewer

Azure OAuth App Configuration

Register a new Azure OAuth app with the details given below –

Homepage URL: FLOWSOURCE_BASE_URL

(eg: https://fs.gcp.flowsource.cognizantproducts.com)

Redirect URI: AUTHUI_BASE_URL/__/auth/handler

(eg: https://auth.fs.gcp.flowsource.cognizantproducts.com/__/auth/handler)

(Note: FLOWSOURCE_BASE_URL and AUTHUI_BASE_URL should be updated with the actual value.)

After the app is created, you should be able to find the Application (client) ID, Application Client Secret and Directory (tenant) ID. You'll need these values in your ALB configuration.

Google IAP Configuration

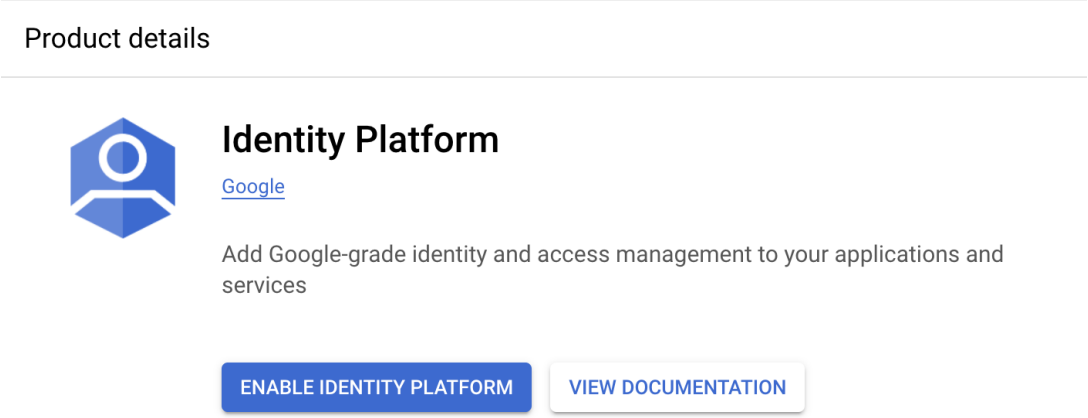
The IAP Configuration involves the following steps –

- Add Identity Provider
- Create Auth UI
- Upload SSL Certificates
- Create ALB for Flowsource app
  - Create health check for backend
  - Set up an SSL certificate resource
- Create ALB for IAP auth UI
  - Create health check for backend
  - Set up an SSL certificate resource
- Configure the firewall rules
- Reserve ALB Frontend IP address
- Add records in Cloud DNS for both ALBs (app & auth UI) frontend IPs
- Configure IAP for the ALB

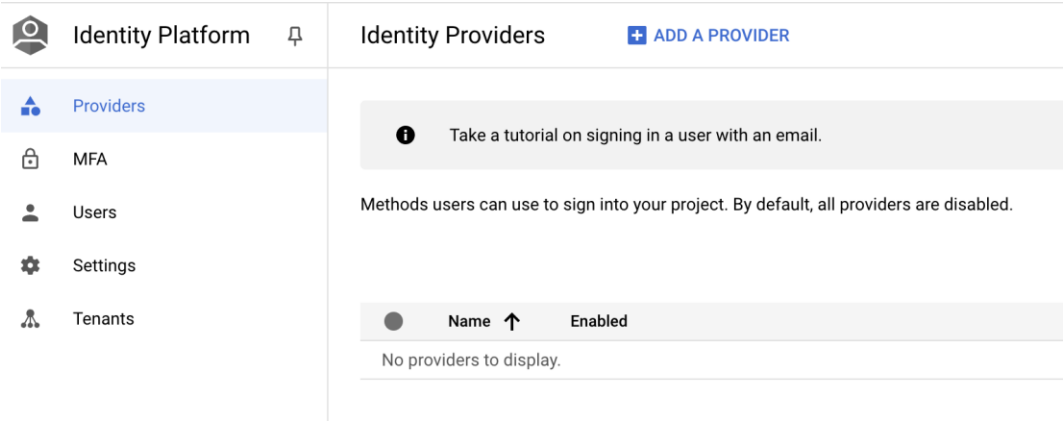
Following are the detailed steps:

Adding an Identity Provider:

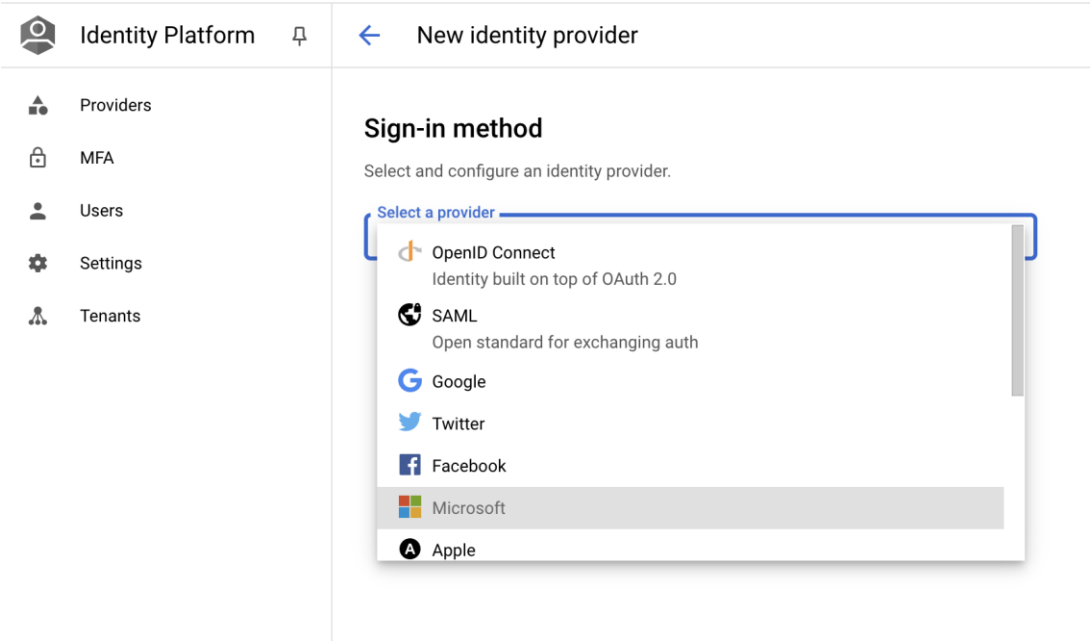
1. Open the Google Identity Providers page – https://console.cloud.google.com/customer-identity/providers
2. Enable Identity Platform (if not enabled already)



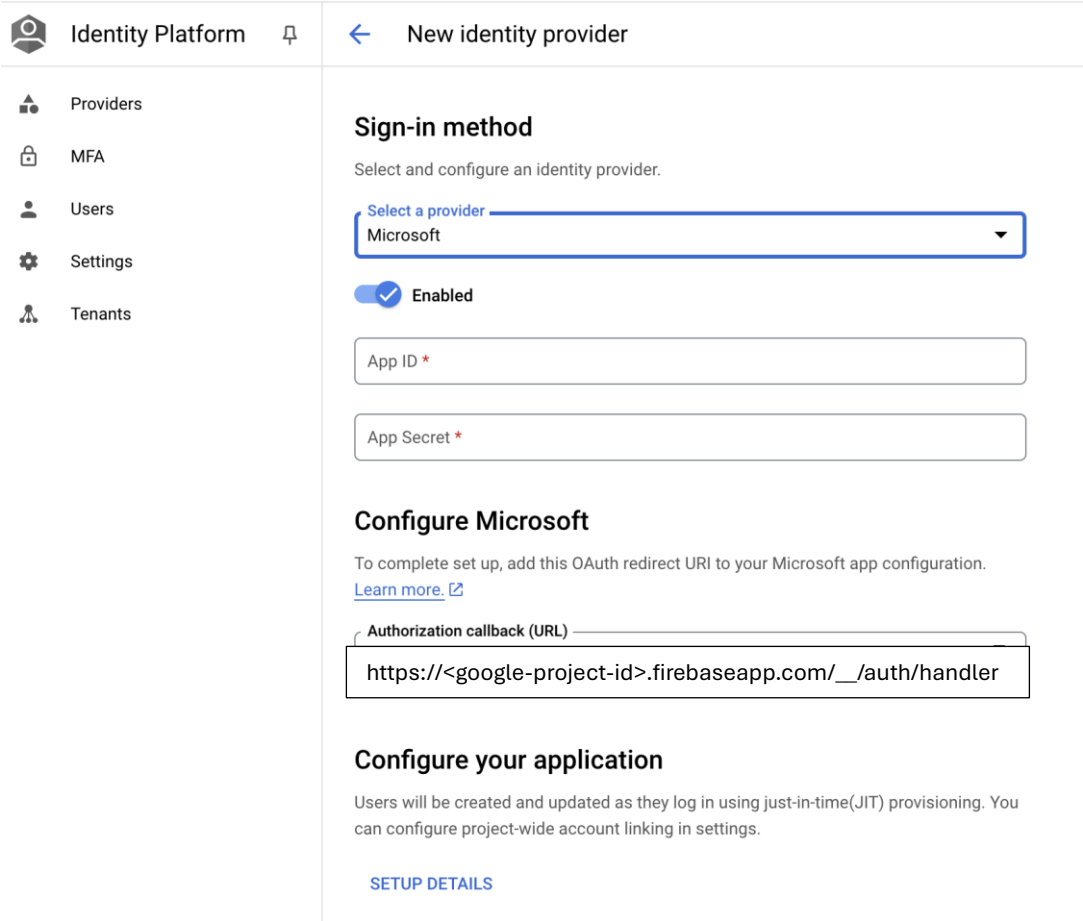
3. Click ADD A PROVIDER



4. Select the Sign-in method as “Microsoft”.

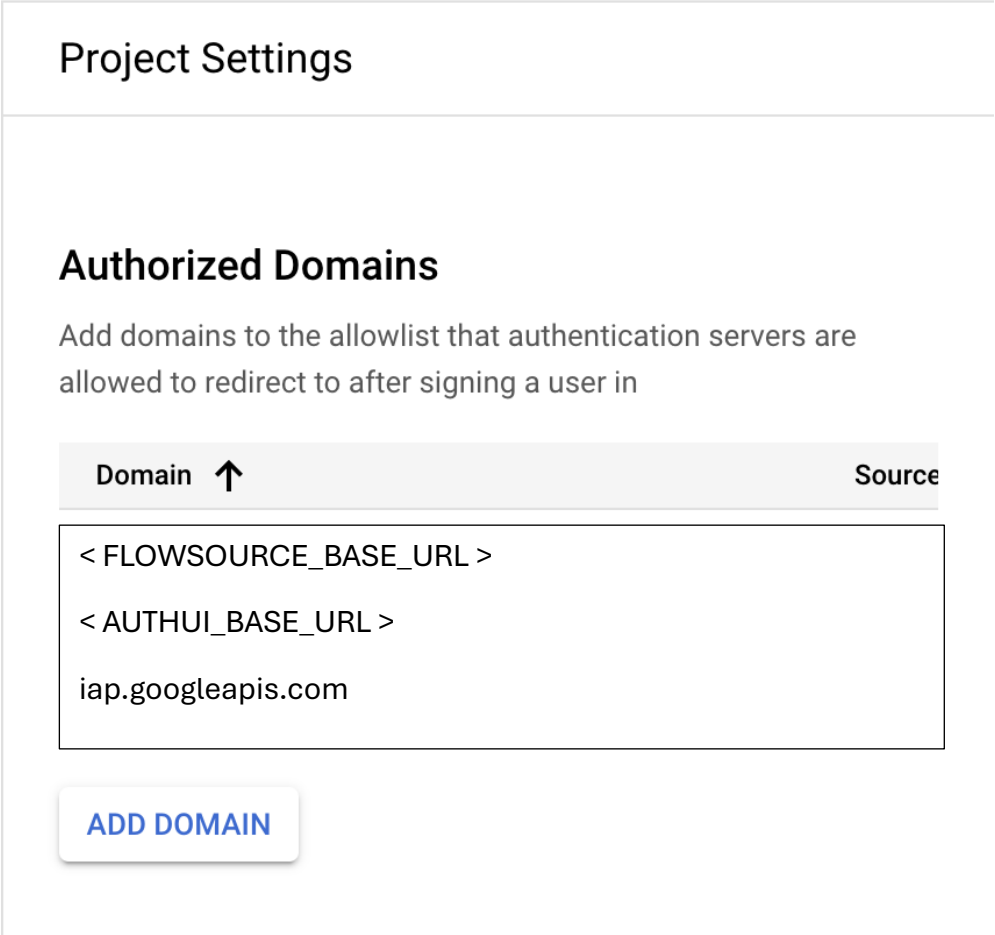


5. Add the **App ID** and **App Secret** from the Microsoft Azure OAuth App.



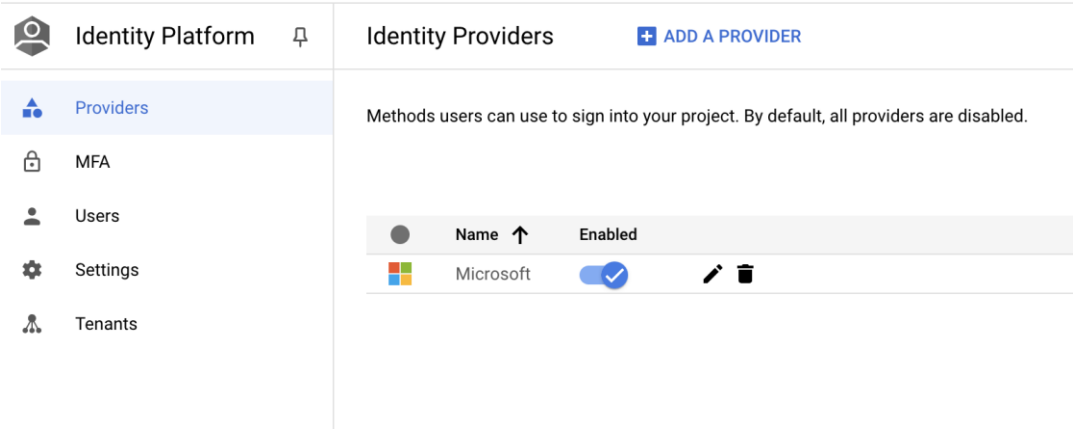
Note: Keep the default value for the **Authorization callback (URL)** and configure this as Redirect URI in Azure OAuth App Configuration.
Example: https://<google-project-id>.firebaseapp.com/__/auth/handler

6. Add the authorized Domains to the Identity Provider.
   Note: The authorized domains should be the DNS of the Flowsource app , the sign-in page and iap.googleapis.com.



   Note: The domain of Flowsource and authui dhould have same valid subdomain (Ex: example.com). Custom domains like .internal will not work with firebase.

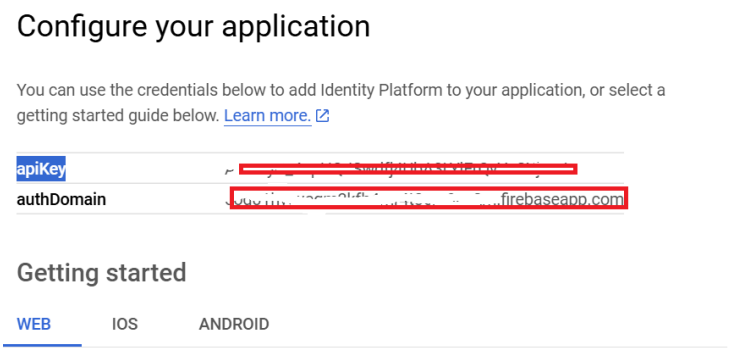
7. Save the configuration. The added Identity Provider will be listed as follows –



Create Auth UI

The Creation of Auth UI is automated, using Terraform. Please follow these steps to deploy an application for the sign-in page on the same GKE Cluster where Flowsource is deployed.

1. Navigate to Identity provider, select the relevant provider and click setup detail to get the "apiKey" value to use as a secret in authui and provide AUTHUI_BASE_URL domain as AUTH_DOMAIN in the values.yaml file.



2. Navigate to the folder "**Flowsource_Package_<release-version>\FlowSource-infra-provision\gcp\gke-flowsource-env\run-on-bastion\authui**"
3. Create a values.yaml at **Flowsource_Package_<release-version>\FlowSource-infra-provision\gcp\gke-flowsource-env\run-on-bastion\authui\values-files** by referring **Flowsource_Package_<release-version>\FlowSource-infra-provision\gcp\gke-flowsource-env\run-on-bastion\authui\authui-charts\docs\example-values.yaml** file.
4. Update the **Flowsource_Package_<release-version>\FlowSource-infra-provision\gcp\gke-flowsource-env\run-on-bastion\authui\terraform.tfvars**

file with relevant details for project_id, region, gke_cluster_name, namespace, values-files location.

5. Update ***backend.tf*** file with the desired bucket name and the key.
6. Run ***terraform init***
7. Run ***terraform plan***
8. Run ***terraform apply***

This will deploy the sign-in page as an application on GKE Cluster.

9. Create a Load Balancer for this service (Please refer below steps to create ALB)
10. Create a DNS record to point to this Load Balancer (Please refer below steps to add records in Cloud DNS)
    Example: auth.< FLOWSOURCE_DNS>

Upload SSL certificates:

Upload the SSL Certificates for <FLOWSOURCE_DNS> and auth.<FLOWSOURCE_DNS> into the certificate Manager.

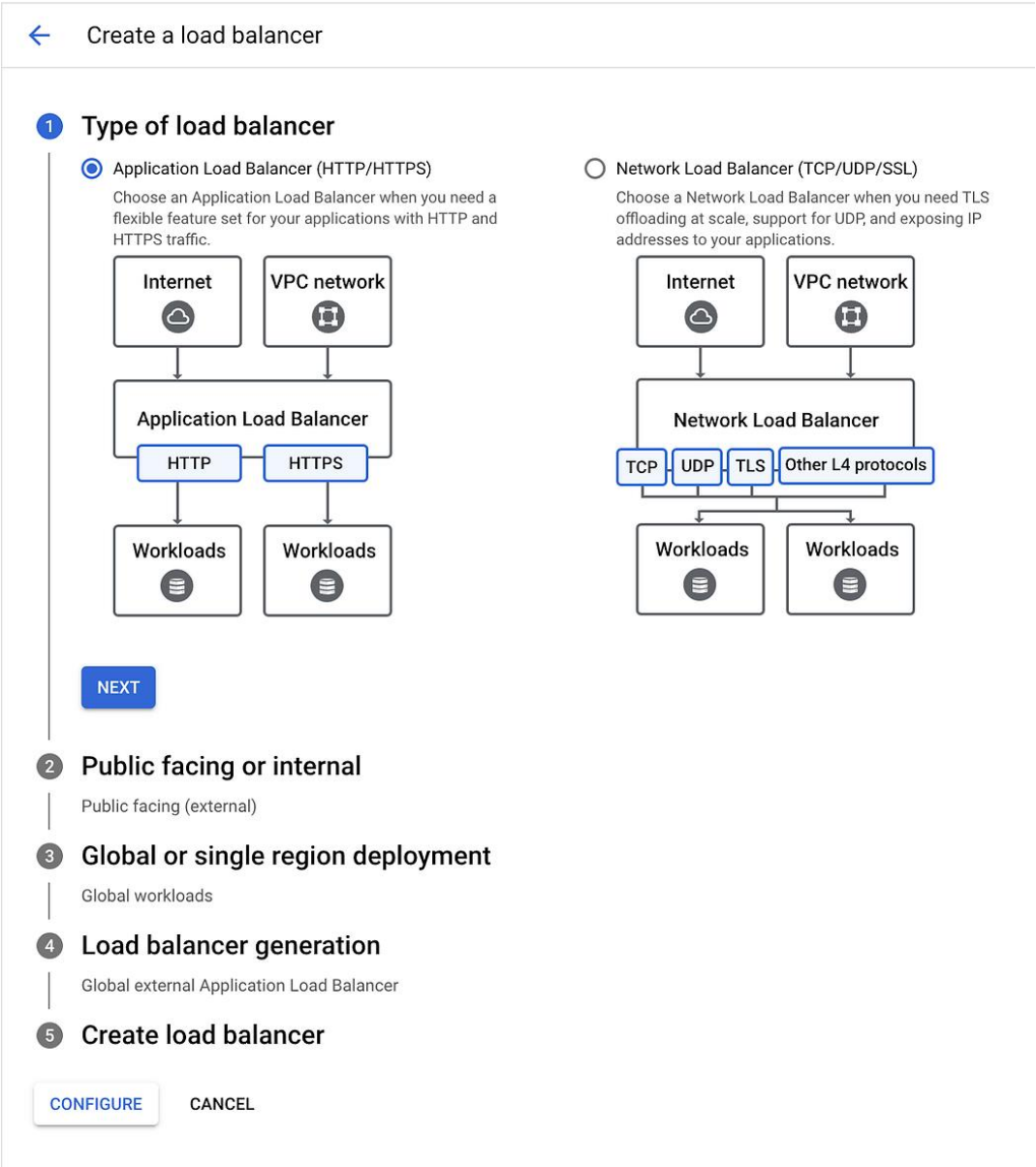
Follow below steps, in case of creation of Self Signed Certificate -

1. Enter name for certificate (e.g., flowsource-gcip-iap-lb-int-app-cert OR flowsource-gcip-iap-lb-int-auth-cert)
2. Execute the below command after replacing <DNS> by your flowsource/auth UI domain name in shell:
   ```
   openssl req -x509 -nodes -days 365 -newkey rsa:2048 -keyout flowsource-iap-app.key -out flowsource-iap-app.crt -subj "/CN=<FLOWSOURCE_DNS>/O=<Organization Name>"
   ```
3. Copy the entire content of .cert file and paste in Certificate* field.
4. Copy the entire content of .key file and paste in Private Key* field.
5. Click Create.

Create ALB for Flowsource app:

This ALB needs to be configured- with backend services for Flowsource Kubernetes cluster instance group in Zonal network endpoint group and Frontend to reserve IP address for https client request for all your backend instances.

1. In the Google Cloud console, go to the **Load Balancing**.
2. Click CREATE LOAD BALANCER



3. Select type → Application Load Balancer (HTTP/HTTPS) and Click Next
4. Select Public facing or internal → Internal and click Next.
5. Select Cross-region or single region deployment → Best for regional workloads and click Next, Configure
6. Enter Load Balancer Name (e.g., flowsource-gcip-iap-lb-int-app)
7. Select Region (in my case it's: us-east-1)
8. Select your VPC name (e.g. **flowsource-vpc**) for the network

9. Click CREATE A BACKEND SERVICE followed by **Backend Configuration**
    A. Enter Name (e.g., flowsource-gcip-iap-lb-int-app-backend)

B. Select Backend type → Zonal network endpoint group, Protocol → HTTP, IP address selection policy → Only IPv4
C. Select a cluster node of Flowsource app in Network endpoint group*
D. Enter Max RPS → 1
E. Click Done

F. Repeat the steps **C** to **E** for all your Flowsource cluster nodes in Network endpoint group*

G. Select Health Check* if already created OR create new one (refer the Steps to create Health Check)

H. Click Create in Create backend service popup and your backend service will be created with 3 network endpoint groups.

10. Click **Frontend configuration**.
    A. Enter name for frontend (e.g., flowsource-gcip-iap-lb-int-app-fe-https)
    B. Select Protocol → HTTPS, Subnetwork → private-subnet-1 (e.g., flowsource-private-subnet-1), Port → 443

    C. Select certificate if available OR create new one (refer the Steps to create SSL certificate),
    D. Select GCP Default for SSL Policy* and Click Done.
    E. Ensure that IP Address for the frontend is reserved. This is necessary for DNS set up.

11. Click **Routing Rule** and configure two Host and path rule for your backend as shown below

Note: enter your flowosource app domain name in Host 2 field.

12. Finally, click CREATE button to create load balancer for flowsource app.

Create ALB for GCIP Auth UI:

This ALB needs to be configured- with backend services for Auth UI Kubernetes cluster instance group in Zonal network endpoint group and Frontend to reserve IP address for https client request for all your backend instances.

1. In the Google Cloud console, go to the **Load Balancing**.
2. Click CREATE LOAD BALANCER
3. Select type → Application Load Balancer (HTTP/HTTPS) and Click Next
4. Select Public facing or internal → Internal and click Next.
5. Select Cross-region or single region deployment → Best for regional workloads and click Next, Configure
6. Enter Load Balancer Name (e.g., flowsource-gcip-iap-lb-int-auth)
7. Select Region (in my case it's: us-east-1)
8. Select your VPC name (e.g., **flowsource-vpc**) for the network

9. Click CREATE A BACKEND SERVICE followed by **Backend Configuration**
    A. Enter Name (e.g., flowsource-gcip-iap-lb-int-auth-backend)
    B. Select Backend type → Zonal network endpoint group, Protocol → HTTP, IP address selection policy → Only IPv4
    C. Select a cluster node of Auth UI in Network endpoint group*
    D. Enter Max RPS → 1
    E. Click Done

    F. Repeat the steps **C** to **E** for all your Auth UI cluster nodes in Network endpoint group*

    G. Select Health Check* if already created OR create new one (refer the Steps to create Health Check)

    H. Click Create in Create backend service popup and your backend service will be created with 3 network endpoint groups.

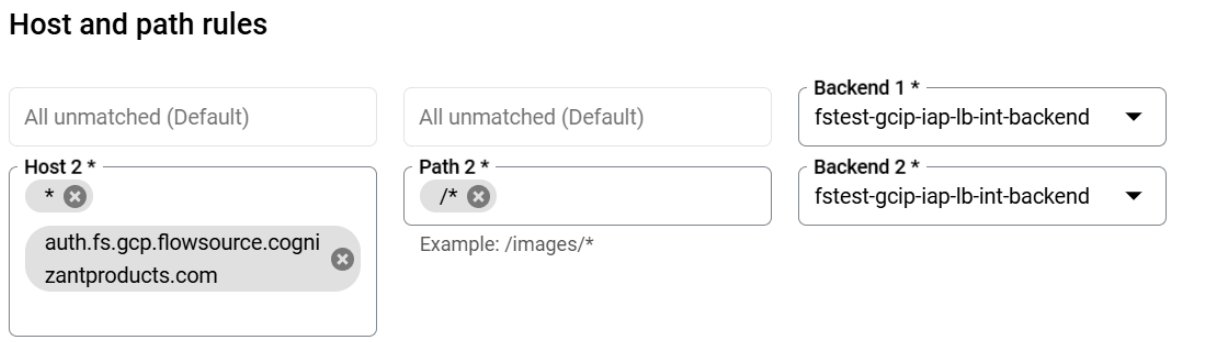
10. Click **Frontend configuration**.
    A. Enter name for frontend (e.g., flowsource-gcip-iap-lb-int-auth-fe-https)
    B. Select Protocol → HTTPS, Subnetwork → private-subnet-1 (e.g., flowsource-private-subnet-1), Port → 443

    C. Select certificate if available OR create new one (refer the Steps to create SSL certificate)

    D. Select GCP Default for SSL Policy* and Click Done.

    E. Ensure that IP Address for the frontend is reserved. This is necessary for DNS set up.

11. Click **Routing Rule** and configure two Host and path rule for your backend as shown below



Note: enter your flowosource auth UI domain name in Host 2 field.

12. Finally, click CREATE button to create load balancer for IAP auth UI.

Create Health Check for Backend services:

1. In the Google Cloud console, go to the **Health checks**.
2. Click CREATE HEALTH CHECK
3. Enter name for Health Check (e.; for flowsource → flowsource-gcip-iap-lb-int-app-hc for flowsource app / **auth UI** → flowsource-gcip-iap-lb-int-auth-hc)
4. Select Regional for Scope.
5. Select Protocol TCP, Enter Port **7007** for **flowsource** app/ Enter Port **8080** for **auth UI**, since auth UI is running on target port 8080.

6. Select your vpc specific region (e.; us-east-1).
7. Click SAVE.

Create the firewall rules for ALB backend services:

This is an ingress rule that allows traffic from the Google Cloud health checking systems (130.211.0.0/22 and 35.191.0.0/16) to reach your backend instances.

1. In the Google Cloud console, go to the **Firewall Rule**.
2. Click CREATE FIREWALL RULE
3. Enter Name (e.g., **allow-healthcheck-ips-to-flowsource)**
4. Select your VPC name (e.g., **flowsource-vpc**) for the network
5. Select "Specified Target tags" for Targets
6. Type your flowsource cluster network tags (e.g., gke-fstest-gke-cluster-fstest-gke-node-pool. Steps: in console, go to the cluster page → select your cluster → click Node tab → click your node in node pool → Metadata → Network tags → copy node pool tag)
7. Select "IPv4 ranges" for Source filter
8. Set Source IPv4 ranges to 130.211.0.0/22, 35.191.0.0/16 and your VPC proxy subnet (In the Google Cloud console, go to the **proxy subnet** → click your-vpc-proxy-only-subnet (e.g.; **fstest-proxy-only-subnet**) to get IP range)
9. Select the TCP checkbox, and then type **7007**, **8080** in the port numbers for ports of Flowsource & Auth UI respectively.
10. Click Create.

Add records in Cloud DNS:

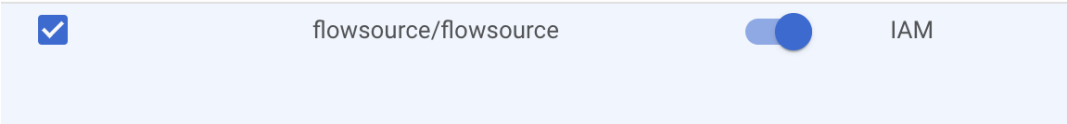
This A record should specify the frontend IP address of the load balancer, ensuring that traffic is routed appropriately.

1. In the Google Cloud console, go to the Cloud DNS page.
2. Click your Zone from Zone List
3. Click Edit → Add Network
4. Select your project
5. Select your VPC (e.g., **flowsource-vpc**) for Network*
6. Click Save
7. Click Add Standard
8. Type your flowsource app DNS name

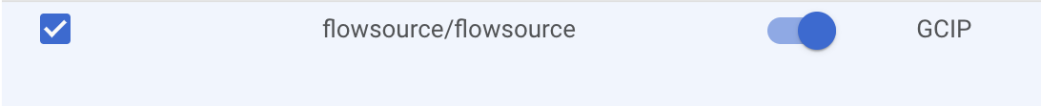
9. Type/ Select Frontend IP of flowsource ALB
10. Click Create.
11. Repeat the same steps **1 to 10** to add auth UI DNS name.

Configure IAP:

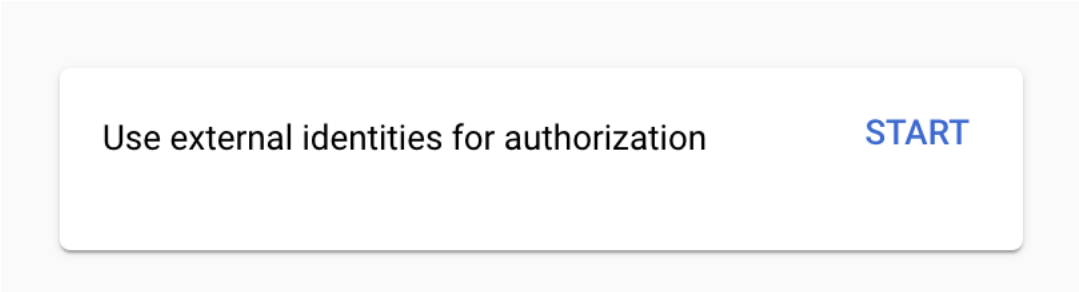
1. Open IAP Page - https://console.cloud.google.com/security/iap
2. Select the Google Project where Flowsource is deployed.
3. Locate the Load Balancer (Backend Service) of the Flowsource application and enable IAP, using the provided switch.



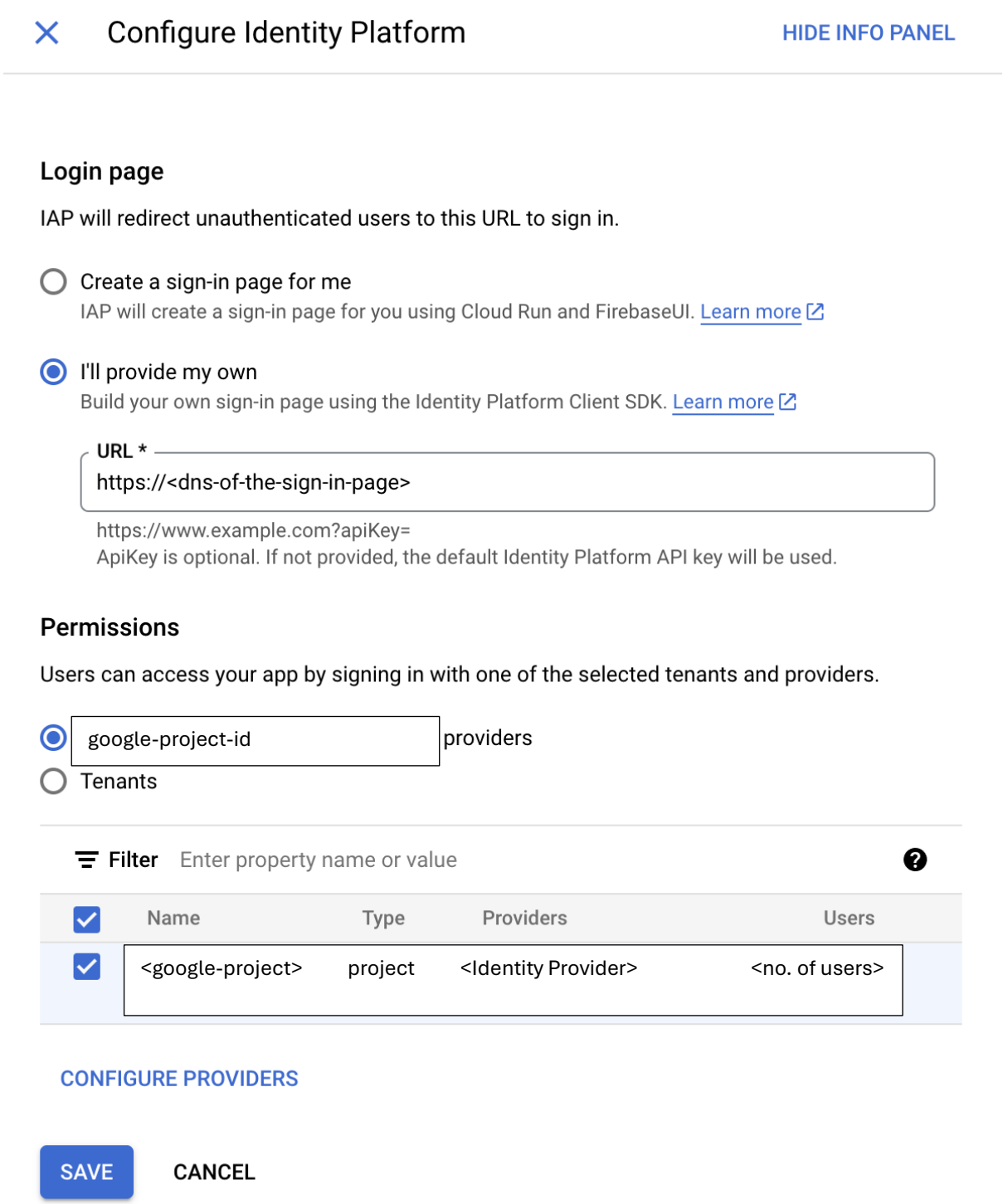
4. The default IAP method is usually IAM. We need to switch that to GCIP with below steps.



5. For attaching the Identity Provider, that was created in the step – Adding an Identity Provider, we need to configure the external identity providers in the "Info Panel" as follows –



6. Update the information for the identity provider as follows –



This completes the IAP configuration.

Flowsource DB Table Updates:

Include the Email to Auth Provider role mapping in the DB “backstage_plugin_auth” in the table “email_to_role_mappings”, for the required admin users.

Command:

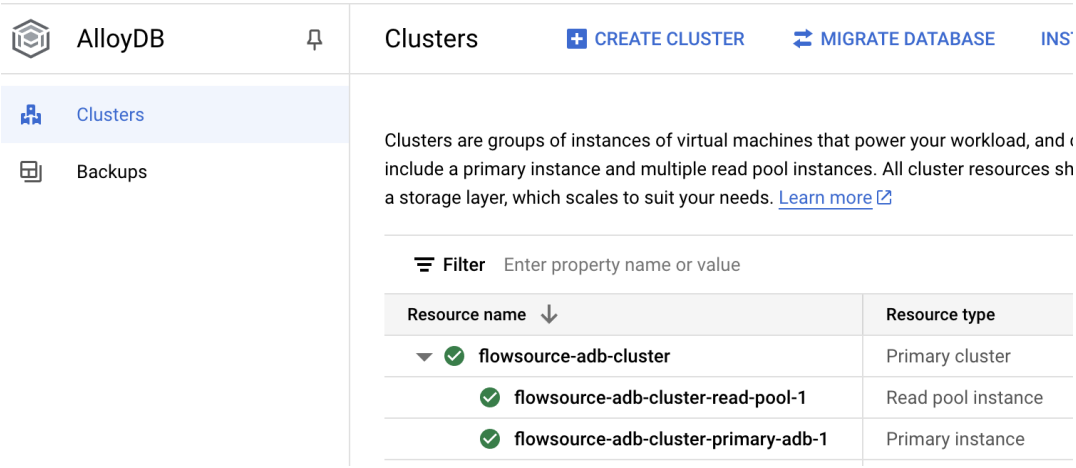
```
INSERT INTO email_to_role_mappings values('user@domain', 'catalog-admin');
```

```
INSERT INTO role_mappings(flowsource_role, auth_provider, auth_provider_role) VALUES ('catalog-admin', 'gcp-iap', 'catalog-admin');
```

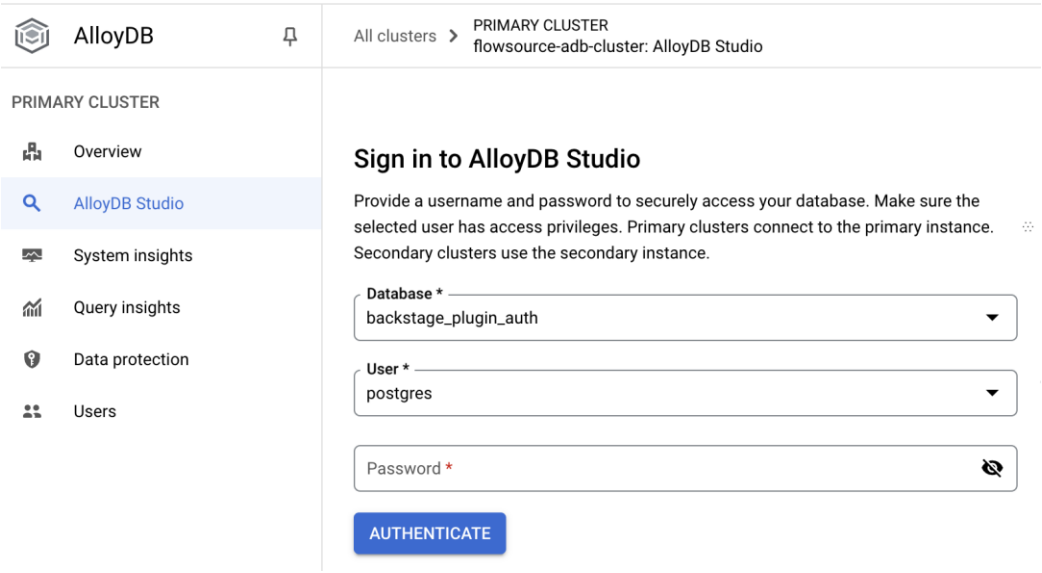
Steps to update the DB –

1. Open Alloy DB Console - https://console.cloud.google.com/alloydb/clusters

2. Click on the Flowsource DB.



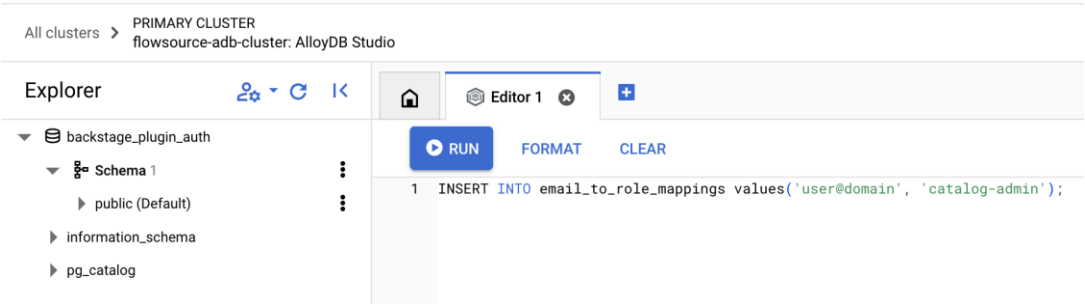
3. Open AlloyDB Studio and login to the DB using the respective credentials.
   Note: Select the Database as “backstage_plugin_auth” and User as “postgres”.



4. Go to the **Editor** tab and run the following command –

   Command:

```
INSERT INTO email_to_role_mappings values('user@domain', 'catalog-admin');
INSERT INTO role_mappings(flowsource_role, auth_provider, auth_provider_role)
VALUES ('catalog-admin', 'gcpip', 'catalog-admin');
```



References:

IAP:
https://cloud.google.com/security/products/iap#documentation

Enabling External Identities:
https://cloud.google.com/iap/docs/enable-external-identities

Creating a custom sign-in page:
https://cloud.google.com/iap/docs/create-custom-auth-ui